

A reanalysis of the FDA’s benefit-risk assessment of Moderna’s mRNA-1273 COVID vaccine stratified not only based on age and sex but also on prior-infection and comorbidity status

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Abstract. We demonstrate that the outcome of a benefit-risk assessment for vaccination may be dramatically impacted by accounting for the benefits derived from prior infection by the disease against which the vaccine is targeted. The United States Food and Drug Administration (FDA) conducted a benefit-risk assessment for Moderna’s COVID vaccine mRNA-1273 prior to its full approval, announced 1/31/2022. The FDA’s assessment focused on males of ages 18–64 years because the agency’s risk analysis was limited to vaccine-attributable myocarditis/pericarditis (VAM/P) given the excess risk among males. The FDA’s analysis concluded that vaccine benefits clearly outweighed risks, even for 18–25-year-old males (those at highest VAM/P risk). We reanalyze the FDA’s benefit-risk assessment using information available through the third week of January 2022 and focusing on 18–25-year-old males. We use the FDA’s framework but extend its model by accounting for protection derived from prior COVID infection, finer age-stratification in COVID-hospitalization rates, and incidental hospitalizations (those of patients who test positive for COVID but are being treated for something else). We also use more realistic projections of Omicron-infection rates and more accurate rates of VAM/P. With hospitalizations as the principal endpoint of the analysis (those prevented by vaccination vs. those caused by VAM/P), our model finds vaccine risks outweighed benefits for 18–25-year-old males, except in scenarios projecting implausibly high Omicron-infection prevalence. Our assessment suggests that mRNA-1273 vaccination of 18–25-year-old males generated between 16% and 63% more hospitalizations from vaccine-attributable myocarditis/pericarditis alone compared to COVID hospitalizations prevented (over a five-month period of vaccine protection assumed by the FDA). The preceding assessment derives from model inputs based on data available at the time of the FDA’s mRNA-1273 assessment. Moreover, these inputs as well as model outputs are validated by subsequently available data.

1. INTRODUCTION

Because vaccines can be given to large populations of healthy persons and can be introduced as mandatory, robust benefit-risk analyses must be conducted to ensure that vaccines have highly favorable benefit-risk profiles [1].

The Food and Drug Administration (FDA) issued emergency-use authorizations for mRNA COVID-19 vaccines from Pfizer (BNT162b2) and Moderna (mRNA-1273) in December 2020 [2]. Once the mRNA COVID-19 vaccines were given to large populations, it became evident that they could cause myocarditis or pericarditis (hereafter myo/pericarditis), inflammation of the heart or of the membrane surrounding the heart, respectively, with studies indicating the highest risk among males younger than 40 years [3, 4, 5, 6, 7, 8, 9].

The FDA has published two benefit-risk analyses of primary series mRNA vaccination, the first for BNT162b2 [10] and the second for mRNA-1273 [11]. The FDA presented its analyses **through scenarios involving different modeling assumptions** to account for **uncertainties in pandemic dynamics** as well as in the benefits and risks of vaccination. To assess vaccine benefits, the FDA **estimated vaccine-preventable COVID-19 cases, hospitalizations, intensive-care unit (ICU) admissions, and deaths** [10, 11]. To assess vaccine risks, the FDA estimated vaccine-attributable myo/pericarditis (VAM/P) cases, hospitalizations, ICU admissions, and deaths [10, 11]. The FDA concluded that the benefits of the mRNA vaccines outweighed VAM/P risk even for males in its highest-risk age-groups: 16–17 years for BNT162b2 [10] and 18–25 years for mRNA-1273 [11].

However, these FDA benefit-risk assessments omitted important inputs in the analysis of both the hospitalizations-prevented benefit of vaccination and VAM/P risk. We focus mainly on the FDA’s analysis for Moderna’s mRNA-1273 to simplify our exposition—there are significant differences in modeling assumptions in the FDA’s Pfizer and Moderna assessments. Also, VAM/P risk is substantially greater for mRNA-1273 than for BNT162b2 [3, 4, 5, 6, 7, 8, 9].

For the FDA’s assessment of mRNA-1273, benefits of vaccination were overestimated. First, the FDA did not account for the **protection conferred by prior infection**; and, at the time of the FDA’s Moderna analysis (mid-January 2022), likely over 70% of 18–25 year-olds in the U.S. had been infected by COVID-19—see Supplement 1. Second, the FDA did not account for incidental COVID-19 hospitalizations. Third, the FDA overestimated benefits for young males by assuming the COVID-hospitalization rate for 18–25 year-olds is equivalent to that for 26–35 year-olds and 36–45 year-olds. Finally, the FDA assumed in its most likely scenario that case and hospitalization rates would remain constant at peak 2021 Omicron levels for 5 months.

By contrast, the FDA underestimated mRNA-1273 VAM/P risks; see, e.g., Table 3 of Section 2.3.

Here, our primary aim is to demonstrate how to incorporate the protection derived from prior infection into a population-level quantitative benefit-risk model for a two-dose vaccination series. Our model has hospitalizations as its endpoint, comparing the hospitalizations-prevented benefit of vaccination to the hospitalization risk attributable to vaccination. We illustrate the utility of our model by applying it to extend and improve the FDA’s mRNA-1273 assessment relative to its hospitalizations endpoint, which can be viewed as the most important of the agency’s endpoints owing to the public-health concern that during waves of the COVID pandemic hospitals might be so overwhelmed that they would not be able to deliver high quality care to their patients.

In addition to prior-infection protection, our modeling accounts for incidental hospitalizations, age-specific infection-hospitalization rates, and a more accurate assessment of VAM/P risk. We apply our model using data available to the FDA at the time of its benefit-risk analysis for mRNA-1273, assumed to be the 3rd week of January 2022 for reasons described in Section 2.1. We present evidence, using real-world data, that our model’s estimates of the hospitalizations-prevented benefit of mRNA-1273 vaccination are significantly more accurate than those the FDA obtained in [11]. We also analyze benefits and risks of vaccination for specific subpopulations such as males 18–25 having prior-infection protection and no comorbidities. **Our model can be adapted to produce for any vaccine designed for deployment on a population-wide scale** a benefit-risk assessment stratified not only based on age and sex but also on prior-infection and comorbidity status.

2. THE FDA’S mRNA-1273 BENEFIT-RISK ASSESSMENT

In [11], the FDA describes its benefit-risk modeling of Moderna’s mRNA-1273 for males ages 18 to 64 years. Outcomes are provided via six scenarios reflecting uncertainties in three major model inputs: COVID-19 incidence, vaccine effectiveness against cases (VE) as well as hospitalization (VEH), and vaccine-attributable myo/pericarditis (VAM/P) rate. At the time of the FDA’s mRNA-1273 analysis, the CDC was recommending a 5-month interval between primary-series completion and a booster dose. Consequently, the FDA evaluated benefits and risks over a 5-month period, which we take to be 1/1/22 – 5/31/22. The FDA assumed that all vaccine effectiveness estimates remained constant over this 5-month *evaluation period*. The FDA’s major modeling assumptions are summarized in Table 1 below.

TABLE 1. The FDA’s mRNA-1273 model-input assumptions for 18–25-year-old males [11, Table 1].

Scenario	COVID-19 Incidence	Vaccine Effectiveness	VAM/P Rate Per Million 2nd doses [11, Table 2]
1	Highest 2021 Incidence, 12/25–12/31	Omicron Dominant; $VE = 30\%$, $VEH=72\%$	128 cases, 110 hospitalizations
2	Average 2021 Incidence	Same as Scenario 1	Same as Scenario 1
3	Lowest 2021 Incidence, 6/5	Same as Scenario 1	Same as Scenario 1
4	Same as Scenario 1	Delta Dominant; $VE = 80\%$, $VEH=90\%$	Same as Scenario 1
5	Same as Scenario 1	Same as Scenario 1	68 cases, 58 hospitalizations
6	Same as Scenario 1	Same as Scenario 1	241 cases, 207 hospitalizations

Using inputs from first row of Table 1 and incidence data from [11, Table 3], the FDA concluded for Scenario 1, its “most likely scenario,” that vaccinating one million males ages 18–25 years with two doses would prevent, over 5 months, 82,484 COVID-19 cases, 4,766 COVID-19 hospitalizations, 1,144 ICU admissions, and 51 deaths due to COVID-19, while risking 128 VAM/P cases, 110 VAM/P hospitalizations, 0 ICU admissions and 0 deaths from VAM/P [11, Table 2]. .

2.1. The Time of the FDA’s mRNA-1273 Analysis. Although the FDA does not define in [11] the time of its benefit-risk analysis for mRNA-1273, a natural choice is the 3rd week of January 2022, which concluded 10 days before the FDA’s announcement of the full approval of mRNA-1273 on 1/31/2022 [12]. This choice is consistent with the FDA’s time-of-analysis definition for its BNT162b2 assessment: the 2nd week of August 2021 [10, Section 2.1], which concluded nine days before the FDA’s announcement of the full approval of BNT162b2 on 8/23/2021 [13]. Consequently, our reanalysis of the FDA’s benefit-risk assessment [11] is based on model inputs derived from information available before 1/22/2022. Subsequent information validating our input choices will often be provided but only via footnotes. Starting in Section 4.2, we discontinue basing our modeling on information only available before 1/22/2022; in Section 4.2, our intent is to show how our model can be modified to incorporate a comorbidity (obesity) as a risk modifier.

2.2. The FDA’s mRNA-1273 Benefits Assessment. The FDA’s computation in [11] of the cases-prevented and hospitalizations-prevented benefits of vaccination is straightforward. For each scenario, CDC data is used to anticipate over the 5-month evaluation period a certain number of COVID cases C and hospitalizations H per million unvaccinated males, and the corresponding evaluation-period benefits per 1 million vaccinations are estimated to be

$$C \cdot VE \text{ cases prevented and } H \cdot VEH \text{ hospitalizations prevented,}$$

where VE and VEH are, respectively, the effectiveness of full mRNA-1273 vaccination against cases and against hospitalization; the FDA assumed that for Omicron $VE = 0.30$ and $VEH = 0.72$, as noted in Table 1 above.

For Scenario 1, its most likely scenario, and two others, the FDA assumed that peak 2021 Omicron-case rates (late December 2021) would be sustained for 5 months, resulting in an exceptionally high projection of COVID-19 cases among the unvaccinated; e.g., $C = 274,947$ for 18–25-year-old males. Similarly, for these scenarios, the FDA assumed that the CDC’s reported late-December 2021 Omicron hospitalization rate for unvaccinated 18–45-year-old males would be sustained for 5 months at approximately 43.54 cases per million daily [11, Table 3] and assumed that the same rate would apply to the younger 18–25 bracket, arriving at $H = 6,619$ hospitalizations. Thus, for these scenarios, the FDA predicted that full mRNA-1273 vaccination of 1 million 18–25-year-old males would prevent $274,947 \cdot 0.30 \approx 82,484$ COVID cases and $6619 \cdot 0.72 \approx 4,766$ COVID hospitalizations over the 5-month evaluation period [11, Table 2].

The following factors contributed to the FDA’s overestimation of vaccine benefits:

- *The FDA did not account for the protection conferred by prior infection.* The FDA’s disregarding prior-infection protection in its mRNA-1273 assessment is difficult to justify. Dan et al. found

“About 95% of subjects retained immune memory at ~ 6 months after [COVID] infection” [14]. A meta-analysis based on studies published December 2020 through August 2021 demonstrated that protection from prior COVID infection “is, at least, equivalent to the protection afforded by complete vaccination of COVID-naïve populations” [15]. The CDC acknowledged in a science brief released 10/29/21 that “The immunity provided by vaccine and prior infection are both high” [16], while another CDC study found that during the Delta wave unvaccinated persons with prior COVID infection had significantly greater protection against infection and hospitalization than did fully vaccinated, COVID-naïve persons [17]. Moreover, in December 2021, the FDA acknowledged, “The omicron variant has significantly more mutations than previous SARS-CoV-2 variants, particularly in its S-gene, the gene that encodes the virus’s spike protein” [18], suggesting that prior infection, entailing exposure to all SARS-CoV-2 proteins (such as the nucleoprotein), would likely yield greater protection against Omicron than would mRNA-1273 vaccination, which entails exposure only to the spike protein of the ancestral strain of SARS-CoV-2. Among the findings of a study of SARS-CoV-2 specific B and T cell memory is “CD8+ T cell responses preferentially target the nucleoprotein, highlighting the potential importance of including the nucleoprotein in future vaccines” [19].¹

Further evidence of the need to account for prior-infection protection includes (i) a CDC estimate that 54.9% of 18–49 year-olds had been infected by COVID-19 by 9/30/2021 [25], (ii) a South African study [26] released 12/14/21 suggesting overall effectiveness of prior-infection protection against Omicron to be about 47% [Supplement 1, Section 4], and (iii) data from England [27] released 12/22/21 suggested that prior infection among the unvaccinated provides significant protection from Omicron hospitalization [Supplement 1, Section 5].

- *The FDA did not account for incidental COVID-19 hospitalizations.* At the time of its mRNA-1273 assessment, incidental-hospitalization rates were well known to be substantial. On 1/9/22, the director of the CDC stated, “[U]p to 40 percent of the patients who are coming in with COVID-19 are coming in ... with something else” [28].² On 1/10/22, Ontario began publishing incidental rates, with rates for January 8–15 exceeding 45% [30]. There is evidence that incidental rates are especially high for the young. For example, a major study [31] found, “As of May 2021, the cumulative rate for hospitalization associated with SARS-CoV-2 infection was 35.9 per 10,000 children” and “When limiting the analysis to include only patients with COVID-19 who required therapeutic interventions, the hospitalization rate decreased 5.5-fold to 6.5 per 10,000 children,” which suggests an incidental rate of over 80% “for children” (persons 0-17-years-old). See Supplement 1 for additional incidental-hospitalization data.

- *The FDA overestimated the proportion of projected COVID cases requiring hospitalization.* All of the FDA’s projected case-hospitalization rates, which are described in Table 2 below, are over four times the Omicron case-hospitalization rate ($0.58\% \approx 56/9,585$) for the unvaccinated of all ages based on Omicron cases in England tested between 1 December 2021 and 14 December 202 [27, Table 3]; moreover, all are well over twice the case-hospitalization rate (0.92%) for the unvaccinated of ages 18–29 in Ontario 12/14/20 – 12/12/21 [32, Table A1a].³ Based on information available at the time of the FDA’s mRNA-1273 assessment, we expect the COVID-case hospitalization rate for the period 12/14/20 – 12/12/21 to be higher than that for Omicron; e.g., “The risk of hospital admission from emergency departments with Omicron was approximately one-third of that for Delta (Hazard Ratio 0.33, 95% CI: 0.30 to 0.37)” [34, p. 4]; see Supplement 1 for more examples.

¹Consistent with the evidence presented in this paragraph, a Qatari study found prior infection provided 52.2% effective protection against symptomatic Omicron infection and 91% effective protection against “severe, critical, or fatal” Omicron, with corresponding effectiveness estimates for 2 doses of mRNA-1273 being 2.2% and 66.3%, respectively [20, Figure 2C, 2D] (study period 12/23/21 – 2/21/22). Findings from the meta-analyses and systematic reviews [21, 22, 23, 24] provide convincing evidence that prior infection provided greater protection against Omicron during the evaluation period than did two-dose vaccination among the COVID-naïve [Supplement 1, Section 7].

²We searched unsuccessfully for published CDC incidental-COVID-hospitalizations data. Figure 2 of a large U.S. study suggests an incidental rate for 0–18 year-olds typically well above 50% during its study period 3/1/2020–1/31/22 and for those 19–64 an incidental rate above 40% for 1/1/2022–1/31/2022 [29].

³The FDA case-hospitalization rates in Table 2 are all well over 5 times the Omicron rate 0.42% for the unvaccinated of all ages reported in Table S10 of [33] (data from 12/15/21 – 1/17/22).

Furthermore, we expect the hospitalization rate for those 18–25 to be lower than that for 18–29 [35]. Factors contributing to the FDA’s overestimating case-hospitalization rates for the 18–25 age range include (i) assuming hospital-admission rates were uniform for ages 18–45, contradicting CDC models [36], and (ii) including incidental COVID hospitalizations. Our source for the 0.58% rate mitigated the incidental-hospitalization issue [27, p. 10], while our source [32] for the 0.92% rate did not.

TABLE 2. The FDA’s case-hospitalization-rate projections for unvaccinated 18–25-year-old males over the evaluation period 1/1/22 – 5/31/22 for the Omicron-based scenarios of [11, Table 2].*

Scenario	Projected Cases	Projected Hospitalizations	Projected Case-Hospitalization Rate
1	274,947	6,619	$\frac{6619}{274,947} \approx 2.41\%$
2	89,017	2,900	$\frac{2900}{89,017} \approx 3.26\%$
3	13,010	882	$\frac{882}{13,010} \approx 6.78\%$
5	274,947	6,619	$\frac{6619}{274,947} \approx 2.41\%$
6	274,947	6,619	$\frac{6619}{274,947} \approx 2.41\%$

*Case and hospitalization projections are obtained from Table 2 of [11] by dividing, respectively, cases-prevented estimates by 0.30 and hospitalizations-prevented estimates by 0.72.

- *The FDA assumed in its most likely scenario (as well as Scenarios 5, and 6) that case and hospitalization rates would remain constant at peak 2021 Omicron levels for 5 months.* Throughout the pandemic, the number of COVID infections has always been significantly larger than the corresponding number of reported cases. For the period February 2020 through September 2021, the CDC estimated the number of infections to be four times the corresponding number of reported cases [25]. Moreover, the case-to-infection multiplier likely increased (e.g. owing to increased home testing) during the Omicron-dominant period in the U.S., which started mid-December 2021. If we conservatively assume that the CDC’s 4:1 infection:case ratio holds over the evaluation period, we find the FDA’s evaluation-period case-count projection of 274,947 per 1 million unvaccinated 18–25-year-old males (Scenarios 1, 5, 6) corresponds to $4 \cdot 274,947 = 1,099,788$ infections, well over 1 million. Rounding down, we deem the FDA’s case-total assumption to be equivalent to the assumption that among one million unvaccinated males, there would be, over 5 months, 1,000,000 infections. Having well over one million infections in a group of 1 million 18–25 year-olds over 5 months is implausible. Multiple infections over a five-month period would be unlikely given the high protection from reinfection resulting from recovery from a recent infection, and any reinfections would be offset by individuals who did not become infected at all (e.g., owing to prior-infection protection, limited exposure to infectious persons, or, possibly, genetic factors [37]).⁴ *Thus, for 3 of the FDA’s 5 Omicron-based scenarios, including its most likely scenario, the FDA essentially assumed that every unvaccinated male 18–25 would be infected by COVID over the evaluation period.*

2.3. The FDA’s mRNA-1273 Risks Assessment. The FDA relied on its Biologics Effectiveness and Safety (BEST) System for the VAM/P case rates appearing in Table 1 above and used the CDC’s Vaccine Safety Datalink (VSD) system to estimate the percentage of VAM/P cases requiring hospitalization [11, Section 2.3.2].

VAM/P-case rates vary considerably depending on the case definition employed and the extent of stratification of risk, based on age, sex, vaccine manufacturer and dose, with most cases occurring postdose 2 [9]. For its mRNA-1273 assessment, the FDA four BEST-System data partners reported, with wide confidence intervals, the following myo/pericarditis incidence rates for 18–25-year-old males during the 7-day interval following dose 2: approximately 35, 70, 146, and 256 cases per million second doses [40, p. 12]. The FDA’s BEST System uses health-claims data not validated by a complete review of patients’ medical charts [11, Section 4], and has “inherent limitations, such as small sample sizes and imperfect sensitivity of ICD-10 codes to identify these rare outcomes” [10, Section 4]. We rely on four sources of data to estimate the VAM/P rate for dose 2 among

⁴The preprint [37], posted 6/10/21, was later published [38]. Even direct inoculation with SARS-CoV-2 does not necessarily produce infection: in the challenge study [39] of the 36 inoculated, 16 “remained uninfected.”

18–25-year-old males: (i) Buchan et al. [41] who apply the Brighton Collaboration’s case definition, (ii) Patone et al. [42] who consider only hospitalized cases of myocarditis, (iii) Sharff et al. [43] who apply the CDC’s case definition [44], and (iv) the FDA (with VSD-based adjustments).

The FDA underestimated mRNA-1273 VAM/P risk:

- *The FDA omitted dose-one VAM/P events.* For its mRNA-1273 analysis [11], the FDA computed the evaluation-period benefits of *two doses* of mRNA-1273 without accounting for the VAM/P risk associated with dose 1, potentially biasing the FDA’s analysis in favor of vaccination. In contrast, for BNT162b2-related VAM/P, the FDA added excess myo/pericarditis cases postdose 1 and postdose 2 to approximate the population-level risk of completing 1 million two-dose vaccination series [10, Section 2.3.1]. Hence, for BNT162b2, the FDA did account for 1st-dose VAM/P risk but did not assess the benefits of dose 1 alone during the interval starting with dose 1 and ending at the point of full vaccination, potentially biasing the analysis against vaccination. Because adverse events of vaccination other than VAM/P were disregarded by the FDA and mRNA vaccination might have unknown long-term adverse health impacts, we believe that even in a benefit-risk analysis assessing benefits only postdose 2, the choice to include 1st-dose VAM/P risk is appropriate, being consistent with the first-do-no-harm principle, the FDA’s choice in its BNT162b2 assessment, and the FDA’s choice in the benefit-risk analysis of its BLA (Biologics License Application) Clinical Review Memorandum for mRNA-1273—see Section 2.4. Thus, we include 1st-dose VAM/P risk for mRNA-1273 in our main analysis. In Supplement 2, we conduct a sensitivity analysis showing that accounting for dose 1 benefits does not substantially change our benefit-risk conclusions.

- *The FDA disregarded evidence that VAM/P risk for mRNA-1273 was over twice that for BNT162b2.* In a “Review Memorandum Addendum” [45], dated 11/19/21, the FDA describes multiple sources of evidence that VAM/P risk for mRNA-1273 is significantly greater than that for BNT162b2, including reports from regulatory agencies of five countries, data from Ontario later analyzed by Buchan et al. [41], and data from the CDC: “[T]he most recent data presented by the Center for Disease Control and Prevention (CDC) analyzed from the Vaccine Safety Datalink (VSD) indicates a higher rate of myocarditis with the Moderna COVID-19 Vaccine than the Pfizer-BioNTech COVID-19 Vaccine”

The “most recent data” from the VSD to which the FDA refers was presented at the CDC’s October 2021 ACIP meeting [46]. Adjusted-rate ratios for 18–39-year-old males from the table on page 27 of [46] suggest the 2nd-dose VAM/P rate for mRNA-1273 is 2.26 times that of BNT162b2. Data from other sources suggests higher rate ratios e.g., a 2nd-dose rate ratio of 5.1 was reported by Buchan et al. for males 18–24 [41, p. 5] and of 5.2 was reported by Husby et al. for males 12–39 [47, paragraph after Figure 3]. A study by Patone et al. of myocarditis cases requiring hospitalization within a 28-day risk interval after a vaccine dose found *among males under 40* a rate of 101 excess myocarditis hospitalizations per million 2nd doses of mRNA-1273 and 12 per million 1st doses, with corresponding BNT162b2 rates being 12 per million 2nd doses and 3 per million 1st doses [42, p. 4].

Based on “preliminary findings” [45, p. 2], having “large uncertainty” owing to “small numbers of observed events” [40, p. 16], as well as omitting dose-1 cases, the FDA suggests that for 18–25-year-old males VAM/P risk for mRNA-1273 is comparable to that for BNT162b2: 128 cases per million 2nd doses of mRNA-1273 [11, Table 4] vs. 131 cases per million full vaccinations of BNT162b2 (among 18–24-year-old males) [10, Table 3], with corresponding hospitalization projections of 110 per million 2nd doses of mRNA-1273 and 131 per million full BNT162b2 vaccinations. *Throughout this paper, we assume that VAM/P rates for males 18–24 well approximate those for males 18–25.*

Using the rate ratios and BNT162b2 VAM/P-risk assessment discussed above, we compute in Section 6.1 of Supplement 1 a VAM/P hospitalization rate of 224 per million 2nd doses of mRNA-1273 for males 18–25 years old.

The FDA’s BEST System also provided evidence that VAM/P risk for mRNA-1273 might be over twice that for BNT162b2. An FDA briefing document for a meeting held 10/14/2021 indicates that one of the agency’s BEST datapartners reported a postvaccination myo/pericarditis incidence rate of 283.7 (95% CI 145.2–573.5) cases per million full mRNA-1273 vaccinations (7 day risk window

after each dose) [48, Section 2.3.3]. Applying a background rate of 4 cases per million per week for 18–25-year-old males [Supplement 1, Section 6.3] to each dose, we find the VAM/P case rate corresponding to 283.7 is 275.7 per million full vaccinations. Assuming a case-hospitalization rate of 100%, as in FDA’s BNT162b assessment, we obtain 275.7 VAM/P hospitalizations per million full mRNA-1273 vaccinations vs. 131 per million full BNT162b vaccinations.

• *The FDA’s meta-analysis of its BEST-System data* [11, Section 2.3.1] yielded VAM/P-rate estimates substantially lower than those of major international studies by Buchan et al. [41] and Patone et al. [42], as well as a U.S. study by Sharff et al. [43]⁵. In Table 3 below, we compare the FDA’s projected mRNA-1273 VAM/P case and hospitalization rates to those of the other sources discussed above. (We have excluded VAM/P-rate sources, such as [49], depending on strictly passive surveillance—see Section 6.8 of Supplement 1.)

TABLE 3. Estimated VAM/P Events Per Million 2nd Doses of mRNA-1273 Among Males 18–24 or 18–25 years old.^a Based on Data Available Before 1 January 2022

Description	FDA Scenarios 1–4 (Scenarios 5 & 6) [11, Table 2]	Buchan et al. ^b [41]	Patone et al. [42, 41]	Sharff et al. [43]	Data From FDA & VSD [10, 40, 46]	Weighted Average of Highlighted Data
Cases per million	128 (68 & 241)	302 ^c	269	525	260	301 [†]
Hospitalizations ^d per million	110 (58 & 207)	242	231	452	224	250 [‡]
Number of events on which estimate is based	21 [40, p. 14]	55 [41, Table 4]	23	7 [43, Table 1]	22	Total Events Columns 3–6 107

^a For discussions of the data in columns 3–6, see Section 6 of Supplement 1; *in particular, data in column 4 are estimates based on findings from [42] for males under 40, adjusted using an age-based myocarditis-incidence ratio as well as pericarditis data from [41].* To obtain the column-5 VAM/P case rate of 525 per million we applied the background myo/pericarditis rate suggested by [41].

^b Buchan et al. applied the Brighton Collaboration’s case definition for myo/pericarditisx, which reduced the reported 417 post-mRNA-vaccination study-period cases of myo/pericarditis to 297 [41, p. 5].

^c This estimated VAM/P-case rate is irrespective of the corresponding 1st-dose type. For homologous dosing (1st dose also mRNA-1273), Table 3 of [41] indicates the rate rises to at least 320 per million when the 2nd dose is taken less than 8 weeks after the 1st (with 4–6 weeks being the recommended interval [50]).

^d For its BNT162b2 assessment, the FDA assumed all VAM/P cases were hospitalized [10, Section 2.3.2.2]. For its mRNA-1273 assessment the FDA assumed VAM/P case-hospitalization rates decrease with age, starting with 86% for males 18–25 and ending with 77% for males 36 and over [11, Section 2.3.2.2]. We apply the 86% rate in columns 4–6 (which, for column 5, matches the case hospitalization rate: 6 of the 7 cases were hospitalized). For column 3, Buchan et al. report a case-hospitalization rate postdose 2 of an mRNA vaccine (all ages, males & females) of 79.2% ($\approx 164/207 \times 100\%$) [41, Table 1]; we expect the rate to be higher for the age range 18–24, but, conservatively, choose a hospitalization rate of 80%.

[†] $301 \approx 302 \cdot \frac{55}{107} + 269 \cdot \frac{23}{107} + 525 \cdot \frac{7}{107} + 260 \cdot \frac{22}{107}$

[‡] $250 \approx 242 \cdot \frac{55}{107} + 231 \cdot \frac{23}{107} + 452 \cdot \frac{7}{107} + 224 \cdot \frac{22}{107}$

2.4. A Seventh Benefit Risk Scenario for mRNA-1273. The FDA’s BLA Clinical Review Memorandum [51] for mRNA-1273, completed 1/28/2022, describes a benefit-risk scenario not matching any scenario of the FDA’s published benefit-risk assessment of mRNA-1273 [11]. The projections of the BLA-Memo Scenario [51, Section 4.7] are described in the initial row of data of Table 4 below and compared to the FDA’s scenarios of [11, Table 2] in particular to Scenario 1, the most likely scenario. Differences include the BLA-Memo Scenario’s accounting for 1st-dose VAM/P risk and projecting 1 million full vaccinations would cause 47 ICU admissions (vs. 0 admissions).⁶

⁵The conclusion of [43] includes “We identified additional valid cases of myopericarditis following an mRNA vaccination that would be missed by the VSD’s search algorithm, which depends on select hospital discharge diagnosis codes. The true incidence of myopericarditis is markedly higher than the incidence reported to US advisory committees.”

⁶Subsequent analyses provide further evidence that the FDA may underestimate VAM/P severity in [11]; e.g., a meta-analysis [52] of 23 studies including 854 patients ages 12–20 years found a 92.6% case-hospitalization rate for postvaccination myo/pericarditis and a 23.2% ICU-admission rate.

TABLE 4. FDA scenarios projecting benefit-risk outcomes per million 18–25-year-old males vaccinated with two primary series doses of mRNA-1273.*

Scenario	Benefits (Outcomes Prevented)				VAM/P Risks			
	Cases	Hospitalizations	ICU Visits	Deaths	Cases	Hospitalizations	ICU Visits	Deaths
BLA-Memo	76,362	1,755	421	4	148	128	47	0
1	82,484	4,766	1,144	51	128	110	0	0
2	26,705	2,088	501	48	128	110	0	0
3	3,903	635	152	7	128	110	0	0
4	219,958	5,957	1,430	63	128	110	0	0
5	82,484	4,766	1,144	51	68	58	0	0
6	82,484	4,766	1,144	51	247	207	0	0

* For Scenarios 1–6, the FDA estimated VAM/P risk only for dose 2. Projections for the FDA’s Scenarios 1–6 are from [11, Table 2]. BLA-Memo-scenario projections are from [51, Section 4.7].”

Consistent with the BLA Memo Scenario, the FDA’s 1/31/22 press release [12] announcing the full approval of the mRNA-1273 vaccine indicates some mRNA-1273 VAM/P cases do require ICU care. Both the BLA Memo [51, Section 4.7] and the FDA’s mRNA-1273 benefit-risk assessment [11, Section 2.3.2.2] attribute their contradictory ICU risk estimates to VSD data. The BLA Memo lacks a specific reference, while the FDA in [11] cites a source of VSD data [53] with collection cutoff point 8/21/2021, over four months before completion of the FDA’s mRNA-1273 assessment.

The VSD data from [53] cited by the FDA [11] is scant, reporting only 21 postvaccination myo/pericarditis cases for those 18–29 [53, p. 16]. A comparison of Buchan et al.’s myo/pericarditis data from Ontario [41, Table 4, collected 6/1/21– 9/4/21] to VSD data through 10/9/21 [46, pp. 8, 16] shows that, among males and females ages 18–39, the number of myo/pericarditis cases per million second doses of mRNA-1273 detected by Public Health Ontario is 3.76 times the number detected by the CDC’s VSD system [Supplement 2, Appendix 3].

3. REANALYSIS: METHODS

Recall that for its case and hospitalization endpoints, the FDA’s benefit-risk model has inputs C and H (the numbers of evaluation-period COVID cases and hospitalizations, respectively, in a test population of one million unvaccinated males) as well as VE and VEH (the fractions of these events prevented by full vaccination). These model inputs are combined to project the cases-prevented and hospitalizations-prevented benefits of vaccinating the test population, and these benefits are compared to the model’s corresponding risk inputs, the numbers of VAM/P cases and hospitalizations projected for the 2nd mRNA-1273 dose.

Our model, which has hospitalizations as its endpoint, is algebraic (like the FDA’s) and thus is widely accessible. Our model extends the FDA’s by not only incorporating prior-infection protection but also by being essentially equivalent to the FDA’s model if prior infection is assumed to provide no protection. Because the FDA relied on CDC data to supply projected values for C and H , the FDA’s model **does not explicitly link cases to hospitalizations and similarly doesn’t explicitly link VE and VEH . In contrast, our model does provide these links. For example, it links infections, a key model input replacing reported cases, to hospitalizations via another key input—the infection hospitalization rate IHR .** Additional input variables are described in Section 3.1 below and summarized in Table 5.

Model inputs are combined to project the hospitalizations-prevented benefit of vaccinating a test population, and this projected benefit is compared to the model’s risk input, the projected number of VAM/P hospitalizations expected to occur in the course of achieving a fully vaccinated test population. *In our modeling, “infection” is any COVID infection, symptomatic or asymptomatic.*

Test population. Our model is based on a hypothetical test population of 1 million males 18–25. We assume that the test population is representative, as of 1/1/2022, of all males in this age group in terms of: distribution of ages within the 18–25 range, comorbidities (e.g., obesity, cardiovascular

disease, diabetes), and the percentage having had a prior COVID-19 infection. We project the number of hospitalizations for Omicron infection that will occur in this test population over the evaluation period 1/1/22 – 5/31/22 (a) assuming all members remain unvaccinated throughout the period, and (b) assuming all members of the population are vaccinated throughout the period (all fully vaccinated, 14 days past dose 2, on 1/1/22). The difference—the projection of (a) minus that of (b)—is the hospitalizations-prevented benefit of vaccinating the test population. We call a test population consisting of infection naïve unvaccinated persons a *baseline test population*.

3.1. Model Inputs. *Infection rate I_r .* We base our modeling on I_r , the fraction of a baseline test population of unvaccinated infection-naïve males 18–25 projected to become infected (symptomatic and asymptomatic) over the FDA’s 5-month evaluation period.

Infection-hospitalization rate H_r . We translate infections to projected hospitalizations using an estimated infection-hospitalization rate H_r for unvaccinated, infection-naïve males 18–25 years old. H_r is the fraction of a population of unvaccinated individuals experiencing a primary COVID infection that will progress to severe disease requiring hospitalization. We emphasize that H_r is a sex-specific rate. The FDA provides data in Table 2 of [10] indicating that for the 18–24 age-range H_r for males is significantly less than that for females.⁷

Prior-infection fraction F_{pi} . This is the fraction of the test population having had a COVID-19 infection prior to the evaluation period.

Vaccine effectiveness against infection E_v . This is the fraction by which the baseline infection rate I_r is reduced due to the protection afforded by vaccination; i.e., the infection rate for a fully vaccinated test population is $I_r(1 - E_v)$.

Prior-infection and hybrid effectiveness against infection, E_{pi} and E_h . These are the fractions by which the baseline infection rate I_r is reduced due, respectively, to the protection afforded by prior infection, or to both vaccination and prior infection (hybrid protection).

Hospitalization risk reduction HRR_v , HRR_{pi} , and HRR_h . These are the fractions by which the hospitalization rate H_r is reduced for populations that are protected respectively by vaccination, prior infection, or both (hybrid protection).

VAM/P hospitalizations $H_{VAM/P}$. Our final model input $H_{VAM/P}$ is the projected number of VAM/P hospitalizations expected to occur in attaining a test population of 1 million fully mRNA-1273 vaccinated 18–25-year-old males. To vaccinate fully 1 million individuals with an mRNA COVID-vaccine, more than 2 million doses must be administered (assuming, e.g., that those who experience myo/pericarditis related to dose 1 do not take dose 2), and the adverse-events associated with each of the over 2 million doses reflect the cost to the population of obtaining the benefits of 1 million full vaccinations. In Section 2.3, based on data available before 1/1/22, we obtained an estimated VAM/P hospitalization rate of 250 per million 2nd doses of mRNA-1273 given to 18–25-year-old males. In Section 6.7 of Supplement 1, we discuss VAM/P-hospitalization risk among 18–25-year-old males for dose 1 of mRNA-1273, justifying a choice of 18 VAM/P hospitalizations per million 1st doses, which is the 1st-dose rate of the FDA’s BLA-Memo Scenario—compare the first two rows of Table 4 above. Thus, for all our scenarios, we set $H_{VAM/P} = 268$.⁸

3.2. Combining Model Inputs to Produce Outputs. Recall that we are considering a test population of 1 million males 18–25 over the evaluation period 1/1/22 – 5/31/22. Let U denote the expected number of Omicron hospitalizations occurring in this test population over the 5-month evaluation period assuming its members remain unvaccinated throughout the period. Let V denote the corresponding number of expected hospitalizations assuming the population is vaccinated throughout the period. The difference $U - V$ is the hospitalizations-prevented benefit of vaccination

⁷The data of Table 1 of [54] is consistent with that of [10, Table 2].

⁸In Supplement 1 (Appendix 2, Section 2.4), we show that our VAM/P hospitalization-rate estimate of 268 per million full mRNA-1273 vaccinations among 18–25-year-old males is consistent with VAM/P data of three major studies [5, 7, 8] appearing in 2022 but available only after 1/22/22, showing the data suggests a VAM/P hospitalization rate between 240 and 277 per million full, homologous mRNA-1273 vaccinations among males 18–25.

for the test population, and the outputs U and V can be written in terms of the model inputs defined in the preceding section (thus, our model is deterministic).

In our modeling, we consider only the possibility of hospitalizations associated with test-population members' first infections during the evaluation period. We expect second infections over a 5-month period to be rare; moreover, infection-hospitalization risk will drop significantly after infection [15, 16, 17]. Our decision to analyze hospitalization risk associated with only first infections during the 5-month evaluation period allows us to assume, as did the FDA, that all effectiveness factors contributing to the analysis remain constant over the evaluation period. Each effectiveness value represents average effectiveness over the evaluation period; in this way, our modeling accounts for waning of vaccine and prior-infection protection.

Deriving U . To derive U , we will assume that the test population remains unvaccinated during the evaluation period. As a first step, suppose that the entire test population is infection naïve. Then, in this baseline population, we expect $(1 \text{ m}) \cdot I_r$ infections to develop over 5 months, resulting in $(1 \text{ m}) \cdot I_r \cdot H_r$ hospitalizations, where $1 \text{ m} = 1 \text{ million}$.

Next suppose a fraction F_{pi} of the unvaccinated test population has had a COVID-19 infection prior to the start of the evaluation period. We must account for the protection against infection and hospitalization afforded by prior infection. We accomplish this in two stages, first accounting for protection against infection, then for protection from progression to severe disease for infected individuals. The first stage depends on the model input E_{pi} , while the second stage uses HRR_{pi} .

So, with fraction F_{pi} of the test population previously infected, $(1 - F_{pi}) \cdot (1\text{m})$ are infection-naïve and will respond like the baseline population described above. That is, we expect $(1 - F_{pi}) \cdot (1\text{m}) \cdot I_r$ infections leading to $(1 - F_{pi}) \cdot (1\text{m}) \cdot I_r \cdot H_r$ hospitalizations from this group. Among the remaining $F_{pi} \cdot (1\text{m})$ members of the test population with prior infection, we expect $F_{pi} \cdot (1\text{m}) \cdot I_r \cdot (1 - E_{pi})$ infections resulting in $F_{pi} \cdot (1\text{m}) \cdot I_r \cdot (1 - E_{pi}) \cdot H_r \cdot (1 - HRR_{pi})$ hospitalizations. Thus the total number of hospitalizations expected in the unvaccinated test population is

$$(3.2.1) \quad U = (1 - F_{pi})(1\text{m})I_rH_r + F_{pi}(1\text{m})I_r(1 - E_{pi})H_r(1 - HRR_{pi}).$$

Deriving V . Next we assume that every individual in the test population is fully vaccinated. Since all members of this test population are vaccinated, those with prior infection have hybrid protection. Once again we reason in two stages, first accounting for the protection against infection, then for the protection against hospitalization for infected individuals. Once again, we assume a fraction F_{pi} of the test population has had a prior COVID-19 infection.

Then, in the fully vaccinated test population, the expected number of hospitalizations is $(1 - F_{pi}) \cdot (1\text{m}) \cdot I_r \cdot (1 - E_v) \cdot H_r \cdot (1 - HRR_v)$ for those without prior infection and $F_{pi} \cdot (1\text{m}) \cdot I_r \cdot (1 - E_h) \cdot H_r \cdot (1 - HRR_h)$ for those with hybrid protection. So the total number of hospitalizations in the population is

$$(3.2.2) \quad V = (1 - F_{pi})(1\text{m})I_r(1 - E_v)H_r(1 - HRR_v) + F_{pi}(1\text{m})I_r(1 - E_h)H_r(1 - HRR_h).$$

Hospitalizations prevented by vaccination. As noted above, the difference $U - V$ is the hospitalizations-prevented benefit of vaccination for the test population:

$$(3.2.3) \quad U - V = (1\text{m})I_rH_r \left(1 - (1 - E_v)(1 - HRR_v)(1 - F_{pi}) - F_{pi} \left(1 + (1 - E_h)(1 - HRR_h) - (1 - E_{pi})(1 - HRR_{pi}) \right) \right).$$

We remark that disregarding the protection afforded by prior infection, as the FDA did, is equivalent in our model to assuming that all persons in the test population are infection-naïve, i.e. that $F_{pi} = 0$. With $F_{pi} = 0$ in Equation (3.2.3), its right-hand side is

$$(3.2.4) \quad [\text{Expected Baseline Hospitalizations}] (1 - (1 - E_v)(1 - HRR_v)),$$

which is essentially the FDA's model H·VEH for hospitalizations prevented. (The quantity $1 - (1 - E_v) \cdot (1 - HRR_v)$ of (3.2.4) corresponds to the VEH formula on page 10 of [34], which is the reference the FDA used in [11] for its VEH estimate of 0.72.) Thus, our model for hospitalizations-prevented $U - V$ represents a natural extension of the FDA's model.

3.3. Estimating Values for Model Inputs. Like the FDA, we account for uncertainties in model inputs by adopting a scenario-based assessment. The FDA’s model-input assumptions for its six scenarios of [11] are summarized in Table 1 of Section 2. Observe that the first three scenarios differ only in their COVID-incidence assumptions, with Scenario 1 representing the FDA’s most likely scenario. Our first three scenarios, Scenarios A–C, are reanalyses, respectively, of the FDA’s Scenarios 1–3 because our COVID incidence assumptions are derived from the FDA’s corresponding assumptions. Based on information available at the time of the FDA’s mRNA-1273 analysis, we consider the COVID incidence assumption of our Scenario D more likely to be accurate than those of Scenarios A–C; also, for Scenario D our values for other inputs represent our best estimates. Thus, Scenario D is our most likely scenario. In Scenario E, our final scenario, we assume that prior infection and mRNA-1273 vaccination provide equivalent protection against Omicron, which, for reasons discussed in Section 2.2, we believe to be the minimum reasonable assumption of prior-infection’s protectiveness relative to vaccination’s protectiveness.

Table 5 below summarizes model inputs with footnotes providing some comments on the estimation of their values—see Supplement 1 for details.

TABLE 5. Summary of Input Variables*

Variables for U and V	Range of Values	Scenario A Values
I_r : Baseline-infection rate—fraction of an unvaccinated, infection-naïve test-population of 18–25-year-old males expected to contract an infection over the 5-month evaluation period. Infections need not accumulate at a constant rate	$0 \leq I_r \leq 1$	1
H_r : Baseline infection-hospitalization rate for infection-naïve unvaccinated males 18–25	$0 \leq H_r \leq 0.01$	0.0014
F_{pi} : Fraction of a test population of 1 million unvaccinated males 18–25 having had a COVID-19 infection before the start of the 5-month evaluation period	$0 \leq F_{pi} \leq 1$	0.62
E_{pi} : Effectiveness of prior-infection protection against COVID-19 infection	$0 \leq E_{pi} \leq 1$	0.47
E_v : Effectiveness of vaccination against COVID-19 infection	$0 \leq E_v \leq 1$	0.3
E_h : Effectiveness of hybrid protection against COVID-19 infection	$0 \leq E_h \leq 1$	0.57
HRR_{pi} : Hospitalization-risk reduction for those reinfected and unvaccinated	$0 \leq HRR_{pi} \leq 1$	0.79
HRR_v : Hospitalization-risk reduction for those fully vaccinated and experiencing their first COVID-19 infection	$0 \leq HRR_v \leq 1$	0.67
HRR_h : Hospitalization-risk reduction for those reinfected and fully vaccinated	$0 \leq HRR_h \leq 1$	0.88
$H_{VAM/P}$: Projected number of VAM/P hospitalizations occurring in the course of attaining a test population of 1 million fully mRNA-1273 vaccinated 18–25-year-old males	$H_{VAM/P} \approx 268$	268

*Remarks. (i) Estimates are based on data available before 1/22/22. (ii) E_{pi} , E_v , and E_h are relative to infection-naïve, unvaccinated persons. (iii) HRR_{pi} , HRR_v , and HRR_h are relative to unvaccinated persons experiencing primary infections. (iv) Our Scenario-A value $I_r = 1$ reflects the assumption that all infection-naïve unvaccinated males 18–25 become infected over the evaluation period, consistent with FDA Scenario-1 assumptions (see Section 2.2). (v) We estimate H_r using an IHR model [55] developed via a seroprevalence-based meta-analysis anchored to the ancestral strain of SARS-CoV-2 and available as a preprint in October 2021 [56]. We assume a 40% incidental-hospitalization rate and make sex-based adjustments based on [10, Table 2]. (See [Supplement 1, Section 2]) (vi) $F_{pi} = 0.62$ is the average of (a) the CDC’s estimate that 54.9% of those 18–49 years old had been infected by COVID-19 before 10/1/21 [25] and (b) an estimate derived in Supplement 1 that 69% of unvaccinated male 18–25 year-olds had been infected by 1/1/2022. (vii) For reasons discussed in Supplement 1, for all scenarios, we use $E_v = 0.3$ and $HRR_v = 0.67$; thus, we are assuming a VEH of $1 - (1 - 0.30)(1 - 0.67) \approx 0.77$, which exceeds the FDA’s VEH (viii) Estimates for E_{pi} , E_h , HRR_{pi} , and HRR_h , are based on data from [26] and [27]. $HRR_{pi} = 0.79$ and $HRR_h = 0.88$ combine with $E_{pi} = 0.47$ and $E_h = 0.57$ to yield overall estimates for protection against hospitalization—88.9% for prior infection and 94% for hybrid protection—consistent with those of meta-analyses [21] and [22].

4. REANALYSIS: RESULTS

In Table 6 below, we compare the hospitalizations-prevented benefit of mRNA-1273 vaccination to corresponding VAM/P-hospitalizations risk, with Scenarios A–C being reanalyses of the corresponding FDA Scenarios 1–3.

TABLE 6. Scenarios comparing the hospitalizations-prevented benefit of mRNA-1273 vaccination of a hypothetical test population of 1 million males 18–25 to the corresponding VAM/P hospitalization risk $H_{\text{VAM/P}}$. For the FDA’s Scenarios 1–3, the values of U and V are determined by the FDA’s model described in Section 2.2, while, for Scenarios A–E, the values of U and V are determined by our model described in Section 3. A green background indicates benefit exceeds risk.

Scenario	Description	Assumptions*	U	V	Hospitalizations Prevented Per Million Vaccinations $U - V$	$H_{\text{VAM/P}}$	Benefit Risk Ratio
FDA Scenario 1	Highest 2021 COVID incidence assumed. FDA’s “most likely scenario”	$U = 6619$ $V = 6619(1-0.72) \approx 1853$	6619	1853	4766	110 (Dose 2 only)	43.33
A	Reanalysis of FDA’s Scenario 1	$I_r = 1, H_r = 0.0014,$ $F_{pi} = 0.62, E_{pi} = 0.47,$ $E_v = 0.30, E_h = 0.57,$ $HRR_{pi}=0.79, HRR_v=0.67,$ $HRR_h = 0.88$	629	168	461	268	1.72
FDA Scenario 2	Average 2021 COVID incidence assumed	$U = 2900$ $V = 2900(1-0.72) \approx 812$	2900	812	2088	110 (Dose 2 only)	18.98
B	Reanalysis of FDA’s Scenario 2	$I_r = 0.356, H_r = 0.0014,$ $F_{pi} = 0.62, E_{pi} = 0.47,$ $E_v = 0.30, E_h = 0.57,$ $HRR_{pi}=0.79, HRR_v=0.67,$ $HRR_h = 0.88$	224	60	164	268	0.61
FDA Scenario 3	Lowest 2021 COVID incidence assumed	$U = 882$ $V = 882(1-0.72) \approx 247$	882	247	635	110 (Dose 2 only)	5.77
C	Reanalysis of FDA’s Scenario 3	$I_r = 0.052, H_r = 0.0014,$ $F_{pi} = 0.62, E_{pi} = 0.47,$ $E_v = 0.30, E_h = 0.57,$ $HRR_{pi}=0.79, HRR_v=0.67,$ $HRR_h = 0.88$	33	9	24	268	0.09
D	COVID incidence twice that for the 2nd COVID wave in the U.S. Our “most likely scenario”	$I_r = 0.456, H_r = 0.0014,$ $F_{pi} = 0.69, E_{pi} = 0.47,$ $E_v = 0.30, E_h = 0.57,$ $HRR_{pi}=0.79, HRR_v=0.67,$ $HRR_h = 0.88$	247	68	179	268	0.67
E	Equivalent protection for prior infected and COVID-naïve vaccinated; incidence as in Scenario D.	$I_r = 0.456, H_r = 0.0014,$ $F_{pi} = 0.69, E_{pi} = 0.30,$ $E_v = 0.30, E_h = 0.57,$ $HRR_{pi}=0.67, HRR_v=0.67,$ $HRR_h = 0.88$	300	68	232	268	0.87

* For discussions of “Assumptions,” see Section 3.3 as well as Supplement 1.

We illustrate our model’s application by carrying out Scenario-A computations. Using Scenario A assumptions from Table 6 and our definitions (3.2.1) and (3.2.2) of U and V , we obtain

$$U = (1 - 0.62) \cdot (1\text{m}) \cdot 1 \cdot 0.0014 + 0.62 \cdot (1\text{m}) \cdot 1 \cdot (1 - 0.47) \cdot 0.0014 \cdot (1 - 0.79) \approx 629, \text{ and}$$

$$V = (1 - 0.62) \cdot (1\text{m}) \cdot 1 \cdot (1 - 0.30) \cdot 0.0014 \cdot (1 - 0.67) + 0.62 \cdot 1 \cdot (1 - 0.57) \cdot 0.0014 \cdot (1 - 0.88) \approx 168.$$

Thus, for Scenario A, the hospitalizations-prevented benefit of vaccination is projected to be $U - V = 629 - 168 = 461$ per million full mRNA-1273 vaccinations, roughly one-tenth of the FDA’s Scenario-1 estimate 4,766 [11]. Although Scenario A has a benefit-risk ratio above 1, its assumption that $I_r = 1$ makes this scenario highly unlikely.⁹ For the more likely scenarios B, D, and E, our modeling suggests that mRNA-1273 vaccination of 18–25-year-old males generated between 16% and 63% more hospitalizations from VAM/P alone compared to COVID hospitalizations

⁹Based on seroprevalence and modeling information, we estimate $I_r \leq 0.5$ [Supplement 2, Section 2.3].

prevented. Comparing the projections of FDA’s “most likely scenario” to ours, we find that the FDA’s benefit-risk ratio, 43.33, is over 60 times more favorable than ours, 0.67. Though Scenario D is our most likely scenario, Scenario E is equally important, being a “best-case scenario” for vaccination incorporating the minimum reasonable assumption of prior-infection protection vs. vaccine protection.

In Section 2 of Supplement 2, we carry out sensitivity analyses exploring how modifying model inputs impacts the benefit-risk picture. For example, if we fix our model-input values, except for I_r , at their Scenario-D levels, then benefit-risk equipoise is reached when $I_r \approx 0.685$. We also illustrate how slider-plots can provide insights concerning equipoise questions involving variation of more than one model input.

4.1. Subpopulation Analyses Based on Prior-Infection Status. There is a compelling case for conducting a benefit-risk assessment for the subpopulation of unvaccinated 18–25-year-old males having prior-infection protection because prior infection not only decreases the hospitalizations-prevented benefit of vaccination but also increases the risk of severe adverse events¹⁰ after vaccination [64, 65], with [64] reporting prior infection is “associated with an increased risk of severe side effects leading to hospital care (1.56 [risk ratio] (95%CI: 1.14–2.12)).” Moreover, prior infection may increase the risk of VAM/P [42]—see Appendix 4 of Supplement 2 for details.¹¹

For the subpopulation of 18–25-year-old unvaccinated males having a COVID-infection prior to the evaluation period 1/1/22 – 5/31/22, we can use our model to compute the expected number of COVID hospitalizations in various scenarios simply by setting $F_{pi} = 1$ while leaving all other inputs unchanged. For example, with $F_{pi} = 1$ in Scenario A, our model predicts $U \approx 155.82$ hospitalizations per million and $V \approx 72.24$, yielding a hospitalizations-prevented benefit of 83.58 per million vaccinated, far below VAM/P hospitalization risk (≈ 268 per million vaccinated).

As we discuss in Appendix 3 of Supplement 1, the preceding paragraph’s analysis may be reinterpreted in terms of evaluation-period hospitalization risk for a typical unvaccinated 18–25-year-old male, COVID-infected prior to the evaluation period. For such a male, the total hospitalization risk of the choice to vaccinate is essentially the sum of his probability of VAM/P hospitalization and his probability of having a breakthrough-case COVID hospitalization during the evaluation period. Our analysis is summarized in Tables 7 and 8 below in terms of hospitalization-risk ratios, with ratios < 1 indicating the choice to vaccinate decreases risk and > 1 indicating it increases risk.

TABLE 7. Hospitalization-risk ratios for a typical male 18–25 with prior-infection protection: risk of hospitalization after choosing mRNA-1273 vaccination (VAM/P or severe COVID-breakthrough infection*) divided by risk of COVID-hospitalization after choosing not to vaccinate.

Probability of Infection I_r^{**}	Probability of COVID-19 Hospitalization Without mRNA-1273 Vaccination	Probability of COVID-19 Hospitalization After mRNA-1273 Vaccination	Hospitalization Risk Ratio vaccinated/unvaccinated, with VAM/P Risk 268 in 1 million
1	155.82 in 1 million	72.24 in 1 million	$2.18 \approx \frac{268+72.24}{155.82}$
0.5	77.91 in 1 million	36.12 in 1 million	$3.90 \approx \frac{268+36.12}{77.91}$
0.25	38.96 in 1 million	18.06 in 1 million	$7.34 \approx \frac{268+18.06}{38.955}$

*COVID hospitalization risk is considered over the 5-month evaluation period

** $I_r(1 - E_{pi})$ is the probability the “typical male” (18–25 with prior-infection protection) will become COVID infected over the evaluation period if unvaccinated; e.g., 0.53 if $I_r = 1$

According to Table 7, our modeling suggests that for a typical unvaccinated 18–25-year-old male with prior-infection, the choice to vaccinate created a hospitalization risk over twice that associated with the choice not to vaccinate even if $I_r = 1$ (evaluation-period infection is certain for an infection-naïve male 18–25). Because a “typical male” has an “average comorbidities level,” the information

¹⁰e.g., those described in [57, 58, 59, 60, 61, 62, 63]

¹¹Survey results reported in [66] also suggest that prior infection may increase the risk of VAM/P; e.g., in Table 4, we see that 2.53% of those having a confirmed SARS-CoV-2 infection before dose 1 of mRNA-1273 reported “chest tightness/pain” during the 7-day period after the dose while only 0.54% of those not having a confirmed infection before dose 1 reported these symptoms; corresponding percentages for dose 2 (Table 5) are 1.78% and 0.98%.

in Table 7 is difficult to apply. However, if our typical unvaccinated male of Table 7 also has no comorbidities, then his hospitalization-risk ratios *should exceed those of Table 7*,¹² because risk-ratio denominators should decrease given no comorbidities and there is no evidence that being free of comorbidities reduces VAM/P risk [67].¹³

For an 18–25-year-old male without prior COVID infection, hospitalization-risk ratios become favorable for vaccination:

TABLE 8. Hospitalization-risk ratios for a typical male 18–25 *without* prior-infection protection: risk of hospitalization after choosing mRNA-1273 vaccination (VAM/P or severe COVID-breakthrough infection) divided by risk of COVID hospitalization after choosing not to vaccinate.

Probability of Infection I_r	Probability of COVID-19 Hospitalization Without mRNA-1273 Vaccination	Probability of COVID-19 Hospitalization After mRNA-1273 Vaccination	Hospitalization Risk Ratio vaccinated/unvaccinated, with VAM/P Risk 268 in 1 million
1	1400 in 1 million	323.4 in 1 million	$0.42 \approx \frac{268+323.4}{1400}$
0.5	700 in 1 million	161.7 in 1 million	$0.61 \approx \frac{268+161.7}{700}$
0.25	350 in 1 million	80.85 in 1 million	$1.00 \approx \frac{268+80.85}{350}$

4.2. Subpopulation Analyses Based on Prior-Infection Status and BMI. At this point, we discontinue confining to footnotes discussions and citations of data unavailable at the time of the FDA’s benefit-risk analysis of mRNA-1273.

The FDA acknowledges, “The health condition of individuals may remarkably impact the B-R profile and its evaluation” [11, Section 4]. **Comorbidities such as obesity may triple COVID-hospitalization risk [69]. Our model can produce a benefit-risk analysis based on comorbidity status by appropriately adjusting its infection-hospitalization-rate input H_r .** We illustrate the process in Appendix 3 of Supplement 1, focusing on obesity because evidence suggests that Body Mass Index (BMI) is the single most potent modifier of COVID-19 risk for those under 60 years old [70, 71]. We use an assessment of COVID-hospitalization risk based on BMI that finds risk is minimized when BMI ≈ 23 and that for $23 \leq \text{BMI} \leq 44$ risk is multiplied by a factor of approximately 1.09 (1.08, 1.10) for each unit increase in BMI [72]. The results of our modeling appear in Table 9 below.

TABLE 9. Hospitalization-risk ratios $\left(\frac{\text{risk if vaccinated (VAM/P \& breakthrough)}}{\text{risk if unvaccinated}} \right)$ for males 18–25 based on BMI. Ratios displayed as pairs: “with prior-infection protection, without prior-infection projection.”

Probability of Infection I_r^*	BMI 23–24	BMI 25–29	BMI 30–34	BMI 35–39	BMI 40–44
1	2.66, 0.39	1.86, 0.30	1.29, 0.23	0.92, 0.19	0.68, 0.17
0.5	5.08, 0.66	3.49, 0.48	2.35, 0.35	1.61, 0.27	1.13, 0.22
0.25	9.92, 1.20	6.74, 0.84	4.47, 0.59	2.99, 0.42	2.03, 0.32

*With prior-infection protection, $I_r(1 - E_{pi})$ is the probability of infection over the evaluation period.

Consistent with Tables 7 and 8, Table 9 indicates that for those having BMI in the intermediate range 25–34, prior-infection status determines whether vaccination is beneficial. However, for the those with high BMI (35–44) in a high-infection-risk scenario Table 9 indicates that vaccination becomes beneficial even for those with prior-infection protection.

4.3. Validating the Model. In Supplement 2, we use Ontario COVID-hospitalizations data from the evaluation period 1/1/22– 5/31/22 to show that our model’s projections of Table 6 for the hospitalizations-prevented benefit of vaccination for 18–25-year-old males are much more accurate than those of the FDA [11].¹⁴ We test our model in two ways: (i) showing that it fits Ontario COVID-hospitalization data with inputs at plausible levels for Ontario over the evaluation period

¹²See, e.g., the BMI 23–24 column of Table 9

¹³See also [68, p. 383].

¹⁴We use Ontario data for many reasons; e.g., unlike the CDC, Public Health Ontario chose to publish incidental COVID-hospitalization rates (starting 1/8/2022) [30]; see [Supplement 2, Section 3] for other reasons.

[Supplement 2 Section 8], and (ii) using Ontario COVID-hospitalization data to predict corresponding data for the U.S. [Supplement 2, Section 7]. For (i), we fit our model to Ontario data for the general-population mix of 18–25-year-old male-and-female Ontarians by using *Mathematica* to solve a sequence of least-squares problems depending on the value of H_r . For $H_r = 0.0011$ and 0.0010 , we obtain model inputs in plausible ranges for Ontario that yield a perfect modeling fit and note that these values of H_r are consistent with COVID hospitalization data from the state of Connecticut. A brief discussion of (ii) is provided in the next paragraph.

Based on COVID hospitalization counts from Public Health Ontario as well as corresponding incidental hospitalization rates, we estimate that vaccinating one million 18–25-year-old Ontarians prevented 152 COVID hospitalizations over the evaluation period—see Table S2.5. Making various adjustments e.g., accounting for the better health of Ontarians vs. Americans, we obtain an *upper-bound* estimate that 214 evaluation-period hospitalizations were prevented by one million full COVID vaccinations of 18–25-year-old males in the U.S.—see [Supplement 2, Table S2.7] and the discussion that follows it. Observe that 214 is an upper bound for our projections of Scenarios B–E (see Table 6) but not an upper bound for any of the FDA’s projections.

Ontario data also supports the definitiveness of the results of Table 7 above given the male has no comorbidities, suggesting that 151 in 1 million is an overestimate of his evaluation-period COVID hospitalization risk [Supplement 2, Section 2.3].

5. DISCUSSION

By accounting for prior-infection protection and using evidence-based model inputs, we find in our Scenarios B–E of Table 6 that risks of mRNA-1273 vaccination outweighed benefits for the general population of 18–25-year-old males, assuming hospitalizations are the principal endpoint of analysis. In particular, for our scenarios predicting plausible infection-level increases over the evaluation period (Scenarios B, D, and E), our model suggests that vaccination of 18–25-year-old males generated between 16% and 63% more hospitalizations from vaccine-attributable myo/pericarditis (VAM/P) alone compared to COVID hospitalizations prevented.¹⁵ Recalling that the FDA found, “[T]he benefits of vaccination with the Moderna COVID-19 vaccine clearly outweigh the risks, even among the male adolescent population” [11, Section 4], we have demonstrated that the conclusions of a benefit-risk assessment for vaccination may be dramatically impacted by accounting for the benefits conferred by prior infection. We have also demonstrated in Tables 7–9 how the benefit-risk picture for vaccination may vary based on an individual’s prior-infection and comorbidity status.

Consistent with our model’s finding that for 18–25-year-old males in the U.S. the risks of mRNA-1273 vaccination outweighed the benefits relative to hospitalizations, we note that France suspended mRNA-1273 use on 10/15/2021, making it available again on 11/8/21 but at half-dose and *only for those over 30 years* [73, p. 4].

We emphasize that the cardiac risk of COVID infection is incorporated into our modeling because our estimates of COVID hospitalizations include those for COVID-related cardiac issues. We address here assertions such as “Importantly, the risk of myocarditis after any dose of COVID-19 vaccine, in any population, is substantially lower than the risk of myocarditis after SARS-CoV-2 infection,” which appeared, without supporting references, in an April 2024 editorial of the *European Heart Journal* [74]. Contrary to the editorial’s assertion, Patone et al. find that “In men younger than 40 years old, the number of excess myocarditis events [leading to hospitalization] per million people was higher after a second dose of mRNA-1273 than after a positive SARS-CoV-2 test (97 [95% CI, 91–99] versus 16 [95% CI, 12–18])” [6]. Karlstad et al. also find significantly higher risk of hospitalized myocarditis after dose 2: 183.9 per million 2nd mRNA-1273 doses [5, Table 2] vs. 13.7 per million positive tests [5, eTable 7] among males 16–24. Moreover, these myocarditis incidence rates expressed in cases per million positive tests overstate the risk of myocarditis after

¹⁵The range 16%–63% is consistent with our model-validation work discussed in Section 4.3: recall we estimate that at most 214 evaluation-period hospitalizations were prevented by one million mRNA-1273 vaccinations of 18–25-year-old males in the U.S. [Supplement 2, Table S2.7], and note that our corresponding VAM/P hospitalization projection of 268 is a little over 25% higher than 214.

COVID infection expressed in cases per million COVID infections (because many infections are not confirmed by positive tests, but essentially all infections among those seeking care for myocarditis are confirmed by positive tests) [75]. Consistent with [75], the authors of [76] note, “[T]he attributable risk estimates for COVID-19 [associated myocarditis] used laboratory confirmed cases as the denominator and will be affected by the proportion of all SARS-CoV-2 infections captured by testing, precluding a direct comparison with vaccine-associated attributable risks.”

The strength of prior-infection protection. As noted in Section 2.2, the FDA’s assumption that prior infection would provide no protection against Omicron is contrary to immunological expectations and studies (e.g., [15, 16, 17, 19, 26, 27]) available at the time of the FDA’s mRNA-1273 analysis suggesting that prior-infection protection would likely equal or exceed vaccine protection among the COVID-naïve. As noted in Table 5 footnotes, we have assumed that vaccination’s effectiveness against hospitalization is 77% while that provided by prior infection is 88.9%. A systematic review [24] involving 21 vaccine-effectiveness studies from the period 12/3/2021 – 4/7/2022 reports,

In contrast to vaccine effectiveness against delta severe disease, the majority of vaccine effectiveness estimates for omicron severe disease were below 75%; for example, thirteen (81%) of sixteen vaccine effectiveness estimates within three months of vaccination with the primary series were below 75% (Figure). Moreover, 13 (42%) vaccine-specific estimates fell below 50% at some point in time after vaccination.

In contrast, the meta-analysis [21] estimates prior-infection’s effectiveness against severe Omicron to be over 80% through 6 months after infection, while the meta-analysis [22, Table S2] estimates its effectiveness to be over 88% through 45 weeks after infection, with the former estimate based on studies from 1/1/22–6/1/22 and the latter, on studies published by 9/31/22. In mid-January of 2022, the minimum reasonable assumption the FDA might have made is that prior-infection protection would equal vaccine protection (as in our Scenario E); instead, in all Omicron-based scenarios of [11], the agency ascribed to vaccination 72% effective protection against hospitalization while assuming prior infection offers no protection.

5.1. Recommendations. Our modeling provides a call to action to policy-forming health agencies to employ multi-variable models like ours that can better stratify vaccination recommendations. To protect vaccine acceptance and promote public trust, the FDA must adopt a rigorous framework for benefit-risk assessment of new vaccines and revisit recommendations as new data becomes available. In particular, the framework must incorporate consideration of the protection that prior infection affords against development of severe disease.

We also recommend United States public-health agencies reassess their priorities. Two years after the first cases of COVID were identified in the U.S., the FDA decided to “not investigate the benefits and risks [of mRNA-1273 vaccination] for specific subpopulations, such as those with comorbidities, due to limited information for these populations” [11, Section 4]. Also, public-health agencies had reason to prioritize understanding the effectiveness of prior-infection protection and had evidence that it was likely to be high. For example, in Pfizer’s phase-3 trial for BNT162b2, 567 participants in the placebo group and 526 in the treatment arm had evidence of prior COVID-19 infection. In each of these cohorts, there was only 1 reinfection (or $< 0.2\%$ for both) [77, p. 27].

5.2. Limitations. Our principal aim has been to illustrate how prior-infection protection, as well as comorbidity status, can be integrated into benefit-risk modeling of a two-dose vaccination series. In applying our model to extend and improve the FDA’s mRNA-1273 assessment [11] relative to its hospitalizations endpoint, we have assumed that young males are hospitalized for COVID at a lower rate than young females, as did the FDA in [10] and [11]. Data from [10] and [54] suggests this rate difference is substantial; however, the rates of [10] and [54] include incidental hospitalizations and incidental rates for males may differ from those for females. We have assumed, as did the FDA, that vaccination and prior infection offer essentially equivalent protection for males and females. Also, we have based our assessment of the difference in hospitalization rates between young males and females on data in Table 2 of [10] representing mostly pre-Delta cases. Relative to Omicron,

sex-based differences in hospitalization rates might have changed. However, the rate differences provided by Table 1 of [54], based on two years of data including that of the initial Omicron wave, are consistent with those of Table 2 of [10].

COVID-hospitalizations data from Ontario [Supplement 2] and from Connecticut [Supplement 1, Appendix 1] suggests that our Omicron IHR value 0.0014 for 18–25-year-old males is a reasonable estimate for the Omicron IHR for *all* 18–25 year-olds (general population mix of males and females). Thus, even if we assume males and females in this age range are hospitalized for COVID at roughly equivalent rates, our IHR value of 0.14% for 18–25-year-old males appears to be a good estimate.

In Section 2 of Supplement 1, based on data available before 1/22/22, we obtain an Omicron IHR estimate for 18–25 year-olds (general-population mix of males and females) of 0.22%, not 0.14%. One possible explanation is that we may have underestimated the incidental COVID-hospitalization rate for 18–25 year-olds. Our IHR estimate for 18–25 year-olds of 0.22% results from assuming an incidental Omicron-hospitalization rate of 40%, a rate suggested by [28] and [30] for hospitalizations with COVID among patients of all ages. However, as noted in Section 2.2 (and discussed in detail in Supplement 1), there is strong evidence that incidental COVID-hospitalization rates are higher among the young than among the elderly. In particular, COVID-hospitalization data discussed in Appendix 1 of Supplement 1 suggests an incidental rate of over 70% among 15–24 year-olds in Connecticut during calendar year 2020. Assuming an incidental rate of about 62.2% (instead of 40%) transforms our IHR estimate of 0.22% into the estimate 0.14%.

For a vaccine having high effectiveness against infection that does not quickly wane, vaccination can have an important benefit not captured by our model—providing an enduring reduction in the level of infection in the population that may allow some members to avoid infection entirely through the contribution of vaccination to herd immunity.

Our risk assessment, like the FDA’s, ignores known, rare vaccine-related adverse events other than myo/pericarditis that may require hospitalization. Finally, our risk assessment, like the FDA’s, ignores the possibility that COVID-19 vaccination may have long-term health effects; for instance, the FDA’s package-insert information for Moderna’s mRNA-1273 begins, “SPIKEVAX has not been evaluated for carcinogenic, mutagenic potential, or impairment of male fertility in animals” [78, Section 13].

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Supplements: [Supplement 1](#), [Supplement 2](#)

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